

Optimal Diffractive Focusing of Vortex Beams

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We formulate diffractive ring focusing of paraxial beams carrying orbital angular momentum (OAM) through a finite circular aperture as a variational optimisation problem. For a fixed topological charge ℓ , we determine the real-valued radial pupil that maximises the intensity at a prescribed ring radius and propagation distance under a normalisation constraint on the input field. The resulting problem admits an exact analytical solution: the optimal pupil combines the familiar Fresnel phase structure with an OAM-dependent Bessel radial modulation, and the associated eigenvalue yields a rigorous upper bound on the achievable ring intensity. We further analyse the lens-assisted configuration and derive a simple bound that allows direct comparison between diffractive and refractive vortex focusing. The theory also provides a quantitative framework for evaluating the performance of practical mask designs, including binary Fresnel–Bessel zone plates and amplitude-weighted pupils. Our results establish the vortex-beam analogue of optimal amplitude-only focusing and identify the fundamental limits governing diffractive concentration of structured waves.

Introduction—The ability to concentrate wave energy at a prescribed location, i.e. focusing, lies at the heart of many areas of science and technology. In optics, acoustics, and matter-wave physics, focusing is typically achieved using lenses or mirrors that shape the phase or arrival time of a propagating field so that different portions of the wavefront arrive in phase and at the same time at the desired focal point [1]. Achieving such control requires adjusting optical paths with sub-wavelength precision, which becomes increasingly challenging as the wavelength decreases. In regimes ranging from X-rays to sub-picometer matter waves, suitable materials may interact only weakly with the field, while fabrication tolerances approach fundamental limits [2]. Broadband or ultrashort pulses introduce an additional complication due to dispersion, which can distort the beam’s temporal structure during propagation [3]. Diffractive approaches offer an alternative to conventional refractive focusing that does not rely on refractive elements. Fresnel’s zone plate, consisting of alternating transparent and opaque rings, demonstrated early on that constructive interference of diffracted waves can generate a bright focal spot even when the field is shaped only through spatial amplitude modulation [4]. Since then, diffractive optical elements such as kinoforms, holographic masks and amplitude apertures have become powerful tools for controlling wave propagation [5]. Importantly, these concepts extend beyond light to other wave systems, including electrons and neutrons, where conventional lenses/phase elements are difficult to realise [6]. In such contexts, shaping the field within a finite aperture can provide a practical lens-free method for concentrating wave intensity [7–11]. Despite their widespread use, the fundamental limits of diffractive focusing have only recently been examined from an optimisation perspective. Efremov et al. [12] posed a benchmark problem: among all real-valued, aperture-limited and normalised input fields, determine the pupil function that maximises the intensity at a prescribed propagation distance. Their variational treatment yields an optimal pupil profile together with a rigorous upper bound on the achievable focal intensity. Remarkably, the analysis shows

that conventional Fresnel-zone constructions are not optimal and that properly designed amplitude modulation can enhance the attainable focal intensity by a factor of $(\pi/2)^2$ relative to standard binary masks. These results establish a principled reference point for evaluating diffractive focusing strategies.

A distinct situation arises for waves carrying orbital angular momentum (OAM). Beams with a helical phase structure $e^{i\ell\varphi}$, ℓ is the topological charge and an integer number, and φ is the azimuth angle of polar coordinates, possess a phase singularity on the optical axis, forcing the intensity to vanish at the centre for any non-zero topological charge ℓ [13]. Consequently, the natural focusing objective is no longer the formation of a bright point but rather the concentration of energy onto an annular ring whose radius depends on both the OAM charge and the radial field profile. Vortex beams have attracted considerable attention in recent decades because of their unique interaction with matter and their growing role in applications such as optical trapping, high-resolution microscopy, nonlinear optics and strong-field physics. In particular, maximising the intensity along the characteristic doughnut-shaped ring while preserving the central phase singularity is essential for techniques such as stimulated emission depletion microscopy [14], for nonlinear processes including multiphoton absorption [15, 16], probing materials’ chirality/dichroism, filamentation [17], and high-harmonic generation [18].

Here, we investigate the diffractive focusing of aperture-limited vortex beams. We formulate the problem as an optimisation over *real* radial pupil functions that shape an incoming OAM mode within a finite aperture, with the objective of maximising the intensity at a prescribed ring radius and propagation distance. This choice reflects the natural focusing geometry imposed by the phase singularity of vortex beams. The resulting formulation leads to an integral-kernel eigenvalue problem, with the principal eigenfunction specifying the optimal pupil profile and the corresponding eigenvalue determining the maximum achievable ring intensity. Beyond providing a practical design rule, this framework establishes

a fundamental performance limit for diffractive focusing of vortex beams. In particular, it identifies the pupil structures that maximise energy concentration on the characteristic annular intensity distribution and clarifies the role of radial field shaping in vortex-beam propagation. The results, therefore, provide the vortex-beam analogue of the optimal amplitude-only focusing problem previously established for conventional beams. By doing so, they offer a rigorous benchmark for evaluating diffractive vortex optics and enable direct comparisons with conventional vortex focusing techniques based on refractive elements, spiral phase plates and zone-plate structures. More broadly, our results reveal how the interplay between Fresnel diffraction and the radial structure imposed by OAM determines the ultimate limits of intensity concentration in vortex beams. This perspective provides a unified framework for analysing and designing diffractive focusing schemes for structured waves, with potential implications for applications ranging from nonlinear optics and strong-field physics to high-resolution microscopy and structured matter-wave beams.

Theory — We consider an aperture-limited field carrying a definite OAM value ℓ , whose transverse profile at the input plane can be written as

$$\Psi_0(\rho, \varphi) = \psi_\ell(\rho) e^{i\ell\varphi}. \quad (1)$$

Here $\rho \in [0, 1]$ denotes the dimensionless radial coordinate, within a circular aperture, and $\psi_\ell(\rho)$ is a real radial pupil function that specifies the amplitude distribution across the aperture – in general, $\psi_\ell(\rho)$ is unique for each OAM value. The propagation of such a field in a paraxial regime, where the deBroglie wavelength is smaller than the spatial extension, is given by the paraxial wave equation, which is similar to a free-particle scalar Schrödinger equation, i.e. $-\frac{\hbar^2}{2M} \nabla^2 \Psi(\rho, \varphi; t) = -i\partial_t \Psi(\rho, \varphi; t)$ — here ∇^2 stands for a 2D Laplacian. We adapt the notations in our previous work and introduce the dimensionless variables $\rho = r/R$ and $\tau = \hbar t/(MR^2)$, where R is the radius of the aperture and M is the particle mass (or, equivalently, the effective propagation parameter for optical waves). The paraxial propagation of the field can be expressed in a dimensionless form independent of the specific physical system, see the Methods for more details. The input field carries a definite OAM charge; thus, the azimuthal phase dependence is preserved during propagation. The field, therefore, evolves as $\Psi(\rho, \varphi; \tau) = e^{i\ell\varphi} \Psi(\rho; \tau)$, where the radial component is given by the Fresnel-Bessel propagation integral,

$$\Psi(\rho; \tau) = \frac{(-i)^{\ell+1}}{\tau} e^{i\frac{\rho^2}{2\tau}} \int_0^1 \rho' d\rho' e^{i\frac{\rho'^2}{2\tau}} J_\ell\left(\frac{\rho\rho'}{\tau}\right) \psi_\ell(\rho'), \quad (2)$$

where $J_\ell(\cdot)$ is the Bessel functions of the first kind of the order ℓ (see Appendix for more details). Our goal is to determine the pupil function $\psi_\ell(\rho)$ such that maximises the intensity at a prescribed ring radius ρ^* at a given focusing time τ_f . Physically, this corresponds to concentrating the energy of the vortex beam onto an annulus, which is the natural focusing geometry

imposed by the phase singularity of OAM beams. We, therefore, seek to maximise the functional $J[\psi_\ell] = |\Psi(\rho^*; \tau_f)|^2$, subject to the normalisation constraint on the input field, i.e., $2\pi \int_0^1 \rho d\rho \psi_\ell^2(\rho) = 1$.

Introducing a Lagrange multiplier λ to enforce this constraint leads to the variational functional

$$\mathcal{L}[\psi_\ell] = J[\psi_\ell] - \lambda \left(2\pi \int_0^1 \rho d\rho \psi_\ell^2(\rho) - 1 \right). \quad (3)$$

Using Eq. (2), the objective functional can be expressed as a quadratic form in the pupil function,

$$J[\psi_\ell] = \int_0^1 \rho d\rho \int_0^1 \rho' d\rho' \mathcal{K}_{\ell, \rho^*}(\rho, \rho', \tau_f) \psi_\ell(\rho) \psi_\ell(\rho'), \quad (4)$$

where the real symmetric kernel

$$\mathcal{K}_{\ell, \rho^*}(\rho, \rho', \tau_f) = \frac{1}{\tau_f^2} \cos\left(\frac{\rho^2 - \rho'^2}{2\tau_f}\right) J_\ell\left(\frac{\rho^* \rho}{\tau_f}\right) J_\ell\left(\frac{\rho^* \rho'}{\tau_f}\right), \quad (5)$$

encodes the propagation dynamics and the target focusing radius. Compared to the kernel obtained in Ref. [12], the present expression contains an additional factor $J_\ell(\rho^* \rho / \tau_f) J_\ell(\rho^* \rho' / \tau_f)$ that arises from the radial structure of vortex beams. For $\ell = 0$ and on-axis focusing ($\rho^* = 0$), Eq. (5) reduces to $\mathcal{K}(\rho, \rho') = \frac{1}{\tau_f^2} \cos\left(\frac{\rho^2 - \rho'^2}{2\tau_f}\right)$, thus recovering the kernel of Ref. [12]. Stationarity of Eq. (3) with respect to variations of $\psi_\ell(\rho)$ yields the integral eigenvalue problem

$$\frac{1}{2\pi} \int_0^1 \rho' d\rho' \mathcal{K}_{\ell, \rho^*}(\rho, \rho') \psi_\ell(\rho') = \lambda \psi_\ell(\rho), \quad (6)$$

where the largest eigenvalue λ determines the maximum achievable ring intensity. A key simplification follows from the trigonometric identity $\cos(a - b) = \cos a \cos b + \sin a \sin b$, which allows the kernel to be written as a sum of two separable contributions,

$$\mathcal{K}_{\ell, \rho^*}(\rho, \rho') = \frac{1}{\tau_f^2} \left[f_c(\rho, \rho^*) f_c(\rho', \rho^*) + f_s(\rho, \rho^*) f_s(\rho', \rho^*) \right], \quad (7)$$

where

$$\begin{aligned} f_c(\rho, \rho^*) &= J_\ell\left(\frac{\rho\rho^*}{\tau_f}\right) \cos\left(\frac{\rho^2}{2\tau_f}\right), \\ f_s(\rho, \rho^*) &= J_\ell\left(\frac{\rho\rho^*}{\tau_f}\right) \sin\left(\frac{\rho^2}{2\tau_f}\right). \end{aligned}$$

To proceed, we introduce the Hilbert inner product $\langle g, h \rangle = 1/(2\pi) \int_0^1 \rho d\rho g(\rho) h(\rho)$. Equation (6), then, becomes

$$\frac{1}{\tau_f^2} \left(f_c(\rho, \rho^*) \langle f_c, \psi_\ell \rangle + f_s(\rho, \rho^*) \langle f_s, \psi_\ell \rangle \right) = \lambda \psi_\ell(\rho). \quad (8)$$

The kernel, therefore, maps any pupil function into the two-dimensional subspace spanned by $\{f_c, f_s\}$ given in Eq. (8). Consequently, the eigenfunctions can be sought within this

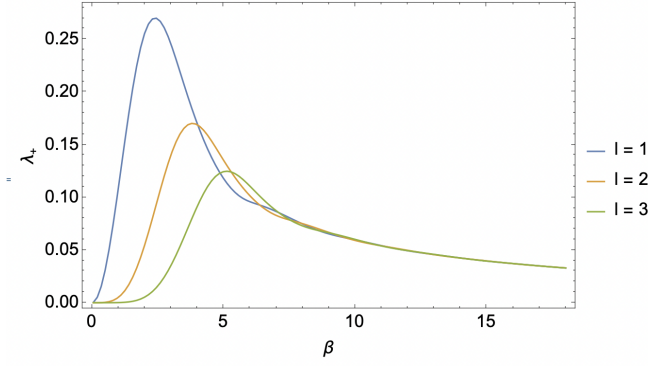


FIG. 1. Continuous optimum focusing performance $\lambda_+(\beta)$ versus the dimensionless radial parameter β for $\ell = 1, 2, 3$ at fixed $\tau_f = 0.25$. The peak shifts to larger β with increasing ℓ , reflecting the outward displacement of the dominant Bessel weight $J_\ell^2(\beta\rho)$ for higher-order OAM modes.

subspace, as $\psi_\ell(\rho) = u f_c(\rho) + v f_s(\rho)$. Defining the overlaps $A = \langle f_c, f_c \rangle$, $B = \langle f_s, f_s \rangle$, and $C = \langle f_c, f_s \rangle$, the eigenvalue problem reduces to the 2×2 matrix equation

$$\frac{1}{\tau_f^2} \begin{pmatrix} A & C \\ C & B \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \lambda \begin{pmatrix} u \\ v \end{pmatrix}. \quad (9)$$

It is convenient to introduce the following integrals

$$\begin{aligned} I_0(\rho^*) &= \frac{1}{2\pi} \int_0^1 \rho d\rho J_\ell^2\left(\frac{\rho\rho^*}{\tau_f}\right), \\ \mathcal{R} &= I_c + iI_s = \frac{1}{2\pi} \int_0^1 \rho d\rho J_\ell^2\left(\frac{\rho\rho^*}{\tau_f}\right) e^{i\frac{\ell^2}{\tau_f}}. \end{aligned} \quad (10)$$

The matrix elements, thus, can be expressed in terms of the above integrals, as $A = (I_0 + I_c)/2$, $B = (I_0 - I_c)/2$, $C = I_s/2$. The eigenvalues of Eq. (9) are therefore

$$\lambda_\pm = \frac{1}{2\tau_f^2} (I_0 \pm |\mathcal{R}|), \quad (11)$$

where $\lambda_+ = (I_0 + |\mathcal{R}|)/(2\tau_f^2)$ corresponds to the maximum achievable ring intensity. Introducing the parameter $\beta = \rho^*/\tau_f$, the corresponding normalised eigenfunctions can be written as,

$$\begin{aligned} \psi_\ell^+(\rho) &= \frac{1}{\sqrt{\tau_f^2 \lambda_+}} J_\ell(\beta\rho) \cos\left(\frac{\rho^2}{2\tau_f} - \theta\right), \\ \psi_\ell^-(\rho) &= \frac{1}{\sqrt{\tau_f^2 \lambda_-}} J_\ell(\beta\rho) \sin\left(\frac{\rho^2}{2\tau_f} - \theta\right), \end{aligned} \quad (12)$$

where the phase shift θ is $\frac{1}{2} \arg(\mathcal{R}) = \frac{1}{2} \arctan(I_s/I_c)$.

Focusing performance of arbitrary pupil functions — The variational solution derived above provides the optimal pupil ψ_ℓ^+ and the corresponding maximum attainable ring intensity λ_+ . In practice, however, experimentally accessible pupils

will in general differ from this optimum. It is, therefore, useful to quantify the focusing performance of an arbitrary normalised pupil relative to the optimal bound. Since the focusing kernel has rank two, its spectral decomposition takes the form

$$\mathcal{K} = \lambda_+ |\psi_\ell^+\rangle\langle\psi_\ell^+| + \lambda_- |\psi_\ell^-\rangle\langle\psi_\ell^-|, \quad (13)$$

where $\psi_\ell^\pm$ are the normalised eigenfunctions obtained in Eq. (12). For any normalised pupil ψ , we define the dimensionless focusing efficiency

$$\eta(\psi) = \frac{J[\psi]}{\lambda_+} = \frac{\langle\psi, \mathcal{K}\psi\rangle}{\lambda_+}, \quad (14)$$

which measures the achieved intensity at the target ring relative to the optimal value. By construction, $0 \leq \eta(\psi) \leq 1$, and $\eta = 1$ is attained only by the optimal pupil ψ_ℓ^+ up to trivial degeneracies.

To make the structure of the efficiency transparent, we decompose an arbitrary normalised pupil as

$$\psi = a_+ \psi_\ell^+ + a_- \psi_\ell^- + \psi_\perp, \quad (15)$$

where $a_\pm = \langle\psi, \psi_\ell^\pm\rangle$, and $\langle\psi_\perp, \psi_\ell^\pm\rangle = 0$. Since the kernel has support only on the active subspace $\text{span}\{\psi_\ell^+, \psi_\ell^-\}$, one has $\mathcal{K}\psi_\perp = 0$. Substituting the decomposition into Eq. (14), immediately gives

$$\eta(\psi) = a_+^2 + \frac{\lambda_-}{\lambda_+} a_-^2. \quad (16)$$

This expression shows that the intensity collected at the target ring depends only on the projection of ψ onto the two-dimensional eigenspace of the focusing operator. Using normalisation, $\|\psi\|^2 = a_+^2 + a_-^2 + \|\psi_\perp\|^2 = 1$, Eq. (16) can be recast as

$$\eta(\psi) = (1 - \|\psi_\perp\|^2) - a_-^2 \left(1 - \frac{\lambda_-}{\lambda_+}\right). \quad (17)$$

Equation (17) makes the two sources of performance loss explicit. The first is leakage outside the active two-dimensional subspace, quantified by $\|\psi_\perp\|^2$; this part of the pupil does not contribute at all to the target-ring intensity. The second is population of the suboptimal eigenmode ψ_ℓ^- , weighted by the penalty factor $1 - \lambda_-/\lambda_+$. Thus, even when a pupil lies entirely within the active subspace, its focusing efficiency is reduced unless it is aligned with the principal eigenfunction. This decomposition also provides a simple measure of robustness against preparation errors. Consider a perturbed input of the form $\psi = \sqrt{1 - \varepsilon^2} \psi_\ell^+ + \varepsilon \psi_\ell^-$, where $\varepsilon \ll 1$, and substiting into Eq. (16) yields $\eta(\psi) = 1 - \varepsilon^2 \left(1 - \frac{\lambda_-}{\lambda_+}\right) + O(\varepsilon^4)$. The intensity loss is therefore quadratic in the admixture, with a coefficient set by the spectral gap. When the two eigenvalues are well separated, contamination by the suboptimal mode produces a correspondingly stronger reduction in performance. By contrast, in the degenerate case $\lambda_+ = \lambda_-$, the quadratic correction vanishes and every normalised state within the

active subspace is equally optimal.

$\ell = 0$ –*Phase Fresnel zones*: As a consistency check, for $\ell = 0$ and $\rho^* = 0$ the optimal amplitude pupil reduces to the once derived in Ref. [12], i.e., $\lambda_{\pm} = \frac{1}{8\pi\tau_f} \left(1 \pm 2\tau_f \left|\sin\left(\frac{\rho}{2\tau_f}\right)\right|\right)$, with the optimal pupil function of $\psi_0^+(\rho) = \frac{1}{\sqrt{4\pi^2\tau_f^2\lambda_+}} \cos\left(\frac{\rho^2}{2\tau_f} - \theta\right)$. We first apply Eq. (17) to the conventional phase Fresnel zone plate in the zero-OAM case. Let ψ_{PFZ} denote the normalized binary pupil

$$\psi_{\text{PFZ}} = 2\sqrt{\pi} \text{sign} \left[\cos \left(\frac{\rho^2}{2\tau_n} - \theta \right) \right], \quad \tau_n = \frac{1}{2\pi n}, \quad (18)$$

where the prefactor is fixed by the normalisation convention used throughout. For $\ell = 0$ and $\tau_n = 1/(2\pi n)$, both eigenstates are equal, $\lambda_+ = \lambda_-$. Thus, the focusing kernel is proportional to the identity on the active subspace. Consequently, every normalized state in $\text{span}\{\psi_0^+, \psi_0^-\}$ yields the same target intensity. In this case, the optimal subspace is spanned by $\psi_0^+(\rho) = \sqrt{8\pi} \cos\left(\frac{\rho^2}{2\tau_n} - \theta\right)$ and $\psi_0^-(\rho) = \sqrt{8\pi} \sin\left(\frac{\rho^2}{2\tau_n} - \theta\right)$.

A direct calculation gives $a_+ = 2\sqrt{2}/\pi$, $a_- = 0$, so that $\eta(\psi_{\text{PFZ}}) = 8/\pi^2$, which is in exact agreement with the results shown in Ref. [12]. This result admits a simple harmonic interpretation. The binary sign operation does not preserve the optimal cosine profile; instead, it redistributes the pupil into higher odd harmonics according to

$$\text{sign}(\cos y) = \frac{4}{\pi} \left(\cos y + \frac{1}{3} \cos 3y + \frac{1}{5} \cos 5y + \dots \right). \quad (19)$$

Only the fundamental harmonic remains in the active two-dimensional eigenspace, whereas the higher harmonics are orthogonal to it and therefore do not contribute to the focal intensity. The efficiency loss of the phase Fresnel zone plate can thus be interpreted as leakage induced by binarisation.

Fresnel–Bessel zones: For nonzero OAM, $\ell \neq 0$, the notion of a Fresnel zone plate is no longer unique. The optimal pupil $\psi_{\ell}^+(\rho)$ contains both the Fresnel oscillation, $\cos\left(\frac{\rho^2}{2\tau_f} - \theta\right)$, and the OAM-dependent radial apodization, $J_{\ell}(\beta\rho)$, where, $\beta = \rho^*/\tau_f$. This suggests two natural generalisations of the conventional binary Fresnel construction.

(i) *Binary Fresnel–Bessel zone mask*. A direct extension is the binary pupil

$$\psi_{\text{FBZ}} = 2\sqrt{\pi} \text{sign} \left[J_{\ell}(\beta\rho) \cos \left(\frac{\rho^2}{2\tau_f} - \theta \right) \right]. \quad (20)$$

Its efficiency depends qualitatively on whether the Bessel factor changes sign within the aperture. A useful reference scale is the first zero $x_{\ell,1}$ of J_{ℓ} . When $\beta < x_{\ell,1}$, the Bessel function does not change sign for $0 \leq \rho \leq 1$. In the small- β regime, one may use $J_{\ell}(\beta\rho) \approx \frac{1}{\ell!} \left(\frac{\beta\rho}{2}\right)^{\ell}$. At the discrete focusing times

$\tau_n = 1/(2\pi n)$, and in the fast-Fresnel regime where the oscillatory phase varies rapidly compared with the Bessel envelope, averaging over the Fresnel oscillations gives

$$\eta_{\text{PFZ}} \approx \frac{8}{\pi^2} \frac{4(\ell+1)}{(\ell+2)^2}. \quad (21)$$

For $\ell = 0$, Eq. (21) reduces to the familiar value $8/\pi^2$, while for $\ell > 0$ the additional factor $\frac{4(\ell+1)}{(\ell+2)^2} < 1$ quantifies the penalty associated with replacing the optimal Bessel apodization by a binary sign profile. This reduction is purely OAM-induced and has no analogue in the zero-charge case.

The behaviour changes once $\beta > x_{\ell,1}$, where the Bessel factor itself develops sign reversals within the aperture. In the large- β regime, the asymptotic form $J_{\ell}(\beta\rho) \approx \sqrt{\frac{2}{\pi\beta\rho}} \cos\left(\beta\rho - \frac{\ell\pi}{2} - \frac{\pi}{4}\right)$ [19] shows that the pupil becomes the sign of a product of two rapidly oscillating factors, one Fresnel-like and one Bessel-like. In this limit, the efficiency approaches

$$\eta_{\text{PFZ}} \approx \frac{512}{9\pi^4}. \quad (22)$$

Thus, once the Bessel oscillations are also binarised, the performance is further reduced relative to the small- β regime.

(ii) *Amplitude-weighted Fresnel–Bessel mask*. A second, less destructive generalisation is to preserve the Bessel apodization while binarising only the Fresnel oscillation. We therefore define

$$\psi_{\text{FBZ}} = \frac{J_{\ell}(\beta\rho)}{\sqrt{I_0(\beta)}} \text{sign} \left[\cos \left(\frac{\rho^2}{2\tau_f} - \theta \right) \right]. \quad (23)$$

This construction retains the radial OAM weighting and therefore isolates the loss associated solely with Fresnel binarisation. Defining $\chi(\rho) = \frac{\rho^2}{2\tau_f} - \theta$, the projections onto the optimal eigenmodes are $a_+^2 = \frac{2A_+^2}{I_0(I_0+|\mathcal{R}|)}$, and $a_-^2 = \frac{2A_-^2}{I_0(I_0-|\mathcal{R}|)}$, with $A_+ = \frac{1}{2\pi} \int_0^1 \rho d\rho J_{\ell}^2(\beta\rho) |\cos \chi(\rho)|$ and $A_- = \frac{1}{2\pi} \int_0^1 \rho d\rho J_{\ell}^2(\beta\rho) \text{sign}[\cos \chi(\rho)] \sin \chi(\rho)$. Substitution into Eq. (16) gives

$$\eta(\psi_{\text{FBZ}}) = \frac{2(A_+^2 + A_-^2)}{I_0(I_0 + |\mathcal{R}|)}. \quad (24)$$

A particularly transparent limit is again the fast-Fresnel regime, in which $J_{\ell}^2(\beta\rho)$ varies slowly over one Fresnel period of $\chi(\rho)$. One may then average over a period to obtain $\langle |\cos \chi| \rangle = \frac{2}{\pi}$, $\langle \text{sign}(\cos \chi) \sin \chi \rangle = 0$, $|\mathcal{R}| \approx 0$. As a result, $A_+ \approx \frac{2I_0}{\pi}$ and $A_- \approx 0$, and thus, Eq. (24) reduces to $\eta(\psi_{\text{FBZ}}) \approx \frac{8}{\pi^2}$. Thus, once the correct Bessel apodization is retained, the efficiency penalty associated with OAM disappears in the fast-Fresnel limit, and the remaining loss is exactly the familiar binary-harmonic penalty already encountered in the $\ell = 0$ case.

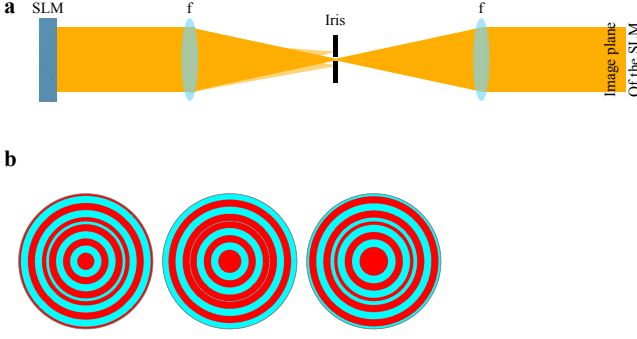


FIG. 2. (a) Schematic of the experimental setup used to generate and analyse the optimal diffractive pupil functions. A linearly polarised Gaussian beam is expanded and modulated by a phase-only spatial light modulator (SLM), which encodes the calculated amplitude kinoforms using an amplitude–phase holographic technique. A $4f$ filtering system selects the desired diffraction order and removes unwanted components, producing the modulated vortex beam. The resulting field propagates in free space and is recorded with a CCD camera at different axial positions. (b) Examples of the calculated pupil masks used to generate ring focusing for topological charges $\ell = 1, 2$, and 3 . The masks implement the optimal radial structure derived in the theory section. (c) Measured transverse intensity distributions during free-space propagation for the corresponding masks, illustrating the formation and evolution of the vortex-ring focus.

Experimental implementation and results — To test the theoretical predictions, we implemented optimal diffractive focusing of vortex beams in the optical domain. The experiment was designed to compare the propagation and ring-focusing performance of different pupil functions, including the binary Fresnel–Bessel mask ψ_{PFBZ} and the amplitude-weighted Fresnel–Bessel mask ψ_{FBZ} , with the corresponding optimal target profile. A linearly polarised Gaussian beam was first expanded to match the active area of a reflective liquid-crystal-on-silicon spatial light modulator (Hamamatsu, 800×600 pixels, pixel pitch $12.5 \mu\text{m}$). The SLM was used to encode the desired real-valued pupil functions by means of an amplitude-phase modulation scheme. In the case of the binary mask ψ_{PFBZ} , the hologram encoded the sign-changing Fresnel–Bessel structure directly. For the amplitude-weighted mask ψ_{FBZ} , the hologram additionally incorporated the required radial apodization associated with the Bessel envelope. In both cases, the calculated pupil functions were converted into phase-only holograms using an amplitude-phase encoding protocol, allowing the target real modulation to be synthesised with the phase-only SLM. After modulation, the first diffraction order was spatially selected using an iris at the Fourier plane of a $4f$ filtering system, thereby removing the unwanted zero- and higher-order diffraction orders generated by the holographic encoding. The selected beam was then re-imaged and allowed to propagate, and its transverse intensity distribution was recorded with a CCD camera at different axial positions. This procedure enabled direct measurement of the propagation dynamics, in particular, the formation of the target annular focus. The experimentally recorded inten-

sity patterns were then compared with the theoretical propagation profiles obtained from Eq. (2). Representative transverse intensity distributions at several propagation planes are shown in the insets of Fig. X, together with the corresponding calculated profiles. In all cases, the observed vortex-ring evolution closely follows theoretical expectations, confirming that the encoded holograms reproduce the intended diffractive pupils with high fidelity. To quantify the focusing performance, we extracted the peak intensity at the target ring as a function of propagation distance and compared the results for the two mask families. The measurements show that the amplitude-weighted Fresnel–Bessel mask ψ_{FBZ} yields a systematically higher ring intensity than the fully binary Fresnel–Bessel mask ψ_{PFBZ} , in agreement with the theoretical efficiency analysis presented above. This difference is consistent with the role of the Bessel apodization factor: when the radial OAM-dependent envelope is retained, the remaining loss is dominated by the binary Fresnel modulation alone, whereas full binarisation introduces an additional penalty associated with the improper radial weighting. Overall, the experimental results validate the central theoretical prediction that optimal diffractive focusing of OAM beams is governed not only by Fresnel phase matching, but also by the Bessel-type radial structure imposed by the vortex geometry. They further show that holographically implemented pupil functions provide a direct and flexible route for testing the fundamental limits of ring focusing in structured beams.

Conclusion— In this work, we have extended the optimal aperture-focusing theory of Ref. [12] to fields carrying non-zero orbital angular momentum. Unlike conventional focusing, vortex beams possess a phase singularity that imposes vanishing on-axis intensity, so the natural focusing objective is to concentrate energy onto a ring rather than a point. As a result, the optimal pupil is no longer determined solely by Fresnel phase matching, but also acquires a radial Bessel structure dictated by the OAM charge and the target ring radius. Despite this additional structure, the optimisation problem remains analytically tractable. We showed that the focusing operator retains an effectively two-dimensional active subspace, which allows the optimal pupil, the maximum achievable ring intensity, and the performance of arbitrary pupil functions to be determined in closed form. This structure provides a natural benchmark for evaluating practical diffractive masks and clarifies how deviations from the optimal pupil degrade the achievable focusing performance. A key outcome of this analysis is that, for $\ell \neq 0$, the limitations of conventional Fresnel-type masks arise from two distinct sources. In addition to the well-known efficiency loss associated with binary-phase quantisation, vortex-beam focusing also requires a specific radial apodization determined by the Bessel envelope. Masks that fail to reproduce this OAM-dependent radial weighting suffer an additional penalty that has no analogue in the $\ell = 0$ case. By contrast, designs that preserve the Bessel modulation while approximating the Fresnel oscillation recover the universal binary-harmonic efficiency limit. The experimental

implementation using spatial light modulation confirms these theoretical predictions and demonstrates that optimal diffractive pupils can be realised and directly compared with practical mask families. More broadly, our results establish the fundamental limits of diffractive focusing for vortex beams and provide a systematic framework for designing high-efficiency ring-focusing elements for structured waves. These ideas are not restricted to optical beams. Because the formulation is expressed entirely in terms of aperture-limited propagation, the same framework applies to a wide class of wave systems, including electron and neutron beams as well as other structured matter waves. The results, therefore, provide a general route for analysing and engineering optimal diffractive focusing in systems where conventional refractive optics is unavailable or impractical.

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Data availability Data from the results presented in this paper may be obtained from the authors upon reasonable request.

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APPENDIX

Derivation of the OAM-projected Fresnel propagator

We start from the dimensionless free Schrödinger (paraxial) equation on the transverse plane $\boldsymbol{\rho} \in \mathbb{R}^2$:

$$i \partial_\tau \Psi(\boldsymbol{\rho}, \tau) = -\frac{1}{2} \nabla_\rho^2 \Psi(\boldsymbol{\rho}, \tau), \quad \Psi(\boldsymbol{\rho}, 0) = \Psi_0(\boldsymbol{\rho}). \quad (25)$$

Taking a spatial Fourier transform

$$\tilde{\Psi}(\mathbf{k}, \tau) = \int_{\mathbb{R}^2} d^2 \boldsymbol{\rho} e^{-i\mathbf{k} \cdot \boldsymbol{\rho}} \Psi(\boldsymbol{\rho}, \tau),$$

the paraxial equation becomes an ODE $i \partial_\tau \tilde{\Psi}(\mathbf{k}, \tau) = \frac{k^2}{2} \tilde{\Psi}(\mathbf{k}, \tau)$ or equivalently $\tilde{\Psi}(\mathbf{k}, \tau) = e^{-ik^2 \tau/2} \tilde{\Psi}_0(\mathbf{k})$. Inverse transforming gives the convolution form

$$\Psi(\boldsymbol{\rho}, \tau) = \int_{\mathbb{R}^2} d^2 \boldsymbol{\rho}' G(\boldsymbol{\rho} - \boldsymbol{\rho}', \tau) \Psi_0(\boldsymbol{\rho}'), \quad (26)$$

where the Green's function is

$$\begin{aligned} G(\boldsymbol{\rho} - \boldsymbol{\rho}', \tau) &= \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} d^2 \mathbf{k} e^{i\mathbf{k} \cdot (\boldsymbol{\rho} - \boldsymbol{\rho}')} e^{-ik^2 \tau/2} \\ &= \frac{1}{2\pi i \tau} \exp\left(\frac{i|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{2\tau}\right). \end{aligned} \quad (27)$$

Hence the full 2D propagated state is

$$\Psi(\boldsymbol{\rho}, \tau) = \frac{1}{2\pi i \tau} \int_{\mathbb{R}^2} d^2 \boldsymbol{\rho}' \exp\left(\frac{i|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{2\tau}\right) \Psi_0(\boldsymbol{\rho}'). \quad (28)$$

Next we convert the propagator to polar coordinates. Write $\boldsymbol{\rho} = (\rho, \varphi)$ and $\boldsymbol{\rho}' = (\rho', \varphi')$. Then $|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2 = \rho^2 + \rho'^2 - 2\rho\rho' \cos(\varphi - \varphi')$. We assume the initial state carries a single OAM charge $\Psi_0(\rho', \varphi') = \psi_\ell(\rho') e^{i\ell\varphi'}$ with $\psi_\ell(\rho') \in \mathbb{R}$. Substituting into the propagator and restricting $\rho' \leq 1$ yields

$$\begin{aligned} \Psi(\rho, \varphi; \tau) &= \frac{e^{i\rho^2/(2\tau)}}{2\pi i \tau} \int_0^1 \rho' d\rho' e^{i\rho'^2/(2\tau)} \psi_\ell(\rho') \\ &\times \int_0^{2\pi} d\varphi' \exp\left(-i\frac{\rho\rho'}{\tau} \cos(\varphi - \varphi')\right) e^{i\ell\varphi'}. \end{aligned} \quad (29)$$

Applying the Jacobi-Anger expansion $e^{-ix \cos \theta} = \sum_{n=-\infty}^{\infty} (-i)^n J_n(x) e^{in\theta}$, the angular integral simplifies to

$$\int_0^{2\pi} d\varphi' \exp\left(-i\frac{\rho\rho'}{\tau} \cos(\varphi - \varphi')\right) e^{i\ell\varphi'} = 2\pi (-i)^\ell e^{i\ell\varphi} J_\ell\left(\frac{\rho\rho'}{\tau}\right), \quad (30)$$

where we used $\int_0^{2\pi} d\varphi' e^{i(\ell-n)\varphi'} = 2\pi \delta_{\ell n}$. Substituting the evaluated angular integral back into the polar-coordinate propagator gives $\Psi(\rho, \varphi; \tau) = e^{i\ell\varphi} \Psi(\rho; \tau)$ with

$$\Psi(\rho; \tau) = \frac{(-i)^\ell}{i\tau} e^{i\rho^2/(2\tau)} \int_0^1 \rho' d\rho' e^{i\rho'^2/(2\tau)} J_\ell\left(\frac{\rho\rho'}{\tau}\right) \psi_\ell(\rho'). \quad (31)$$

If the initial state is not a single ℓ . The above analysis can be expanded to a general initial state $\Psi_0(\rho', \varphi')$ by expanding the initial field in angular harmonics $\Psi_0(\rho', \varphi') = \sum_{m \in \mathbb{Z}} e^{im\varphi'} \Psi_{0,m}(\rho')$. By linearity, each m propagates independently with the same radial kernel in Eq.(31).